

Online Appendix to
Measuring Religion from Behavior:
Violence, Climate Shocks and Religious Adherence in
Afghanistan

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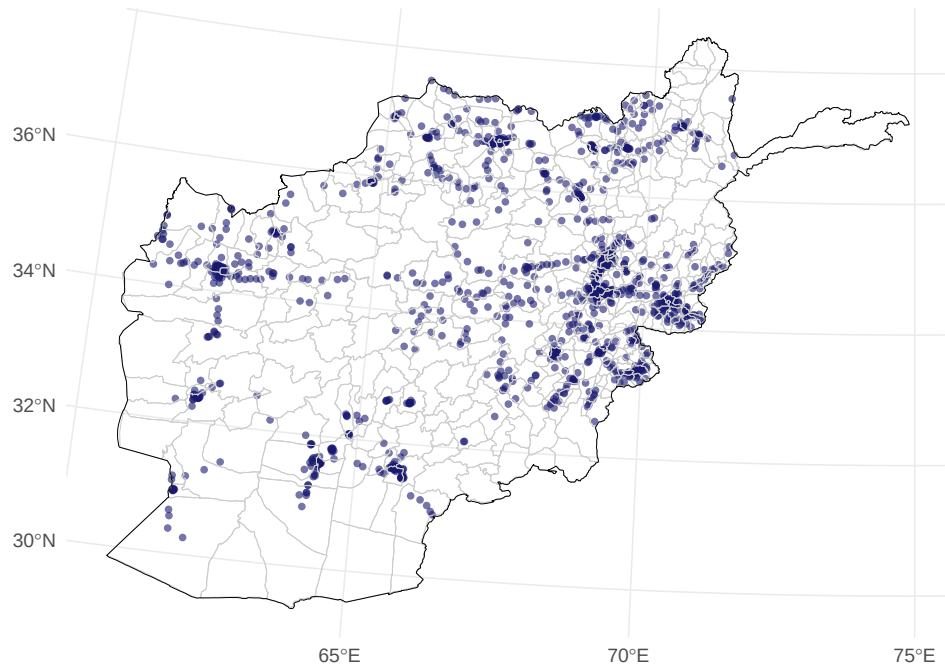
A Appendix figures and tables

Figure A1: Poster Prohibiting Cell Phone Use



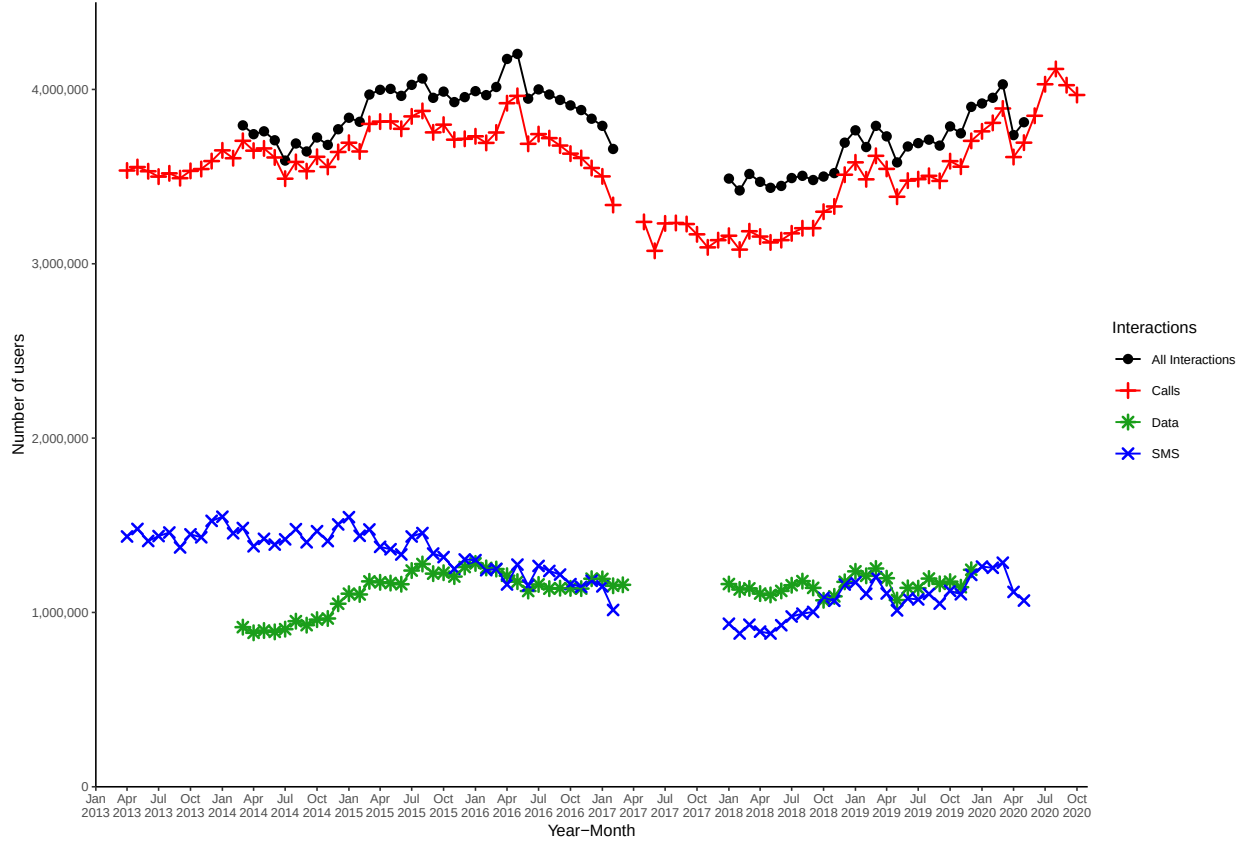
Notes: Poster found in Hounslow Muslim Centre. See [here](#).

Figure A2: Cellphone Towers in our Sample



Notes: This figure shows the location of the cellphone towers in our sample, along with district boundaries in grey.

Figure A3: The Number of Cell Phone Users over Time



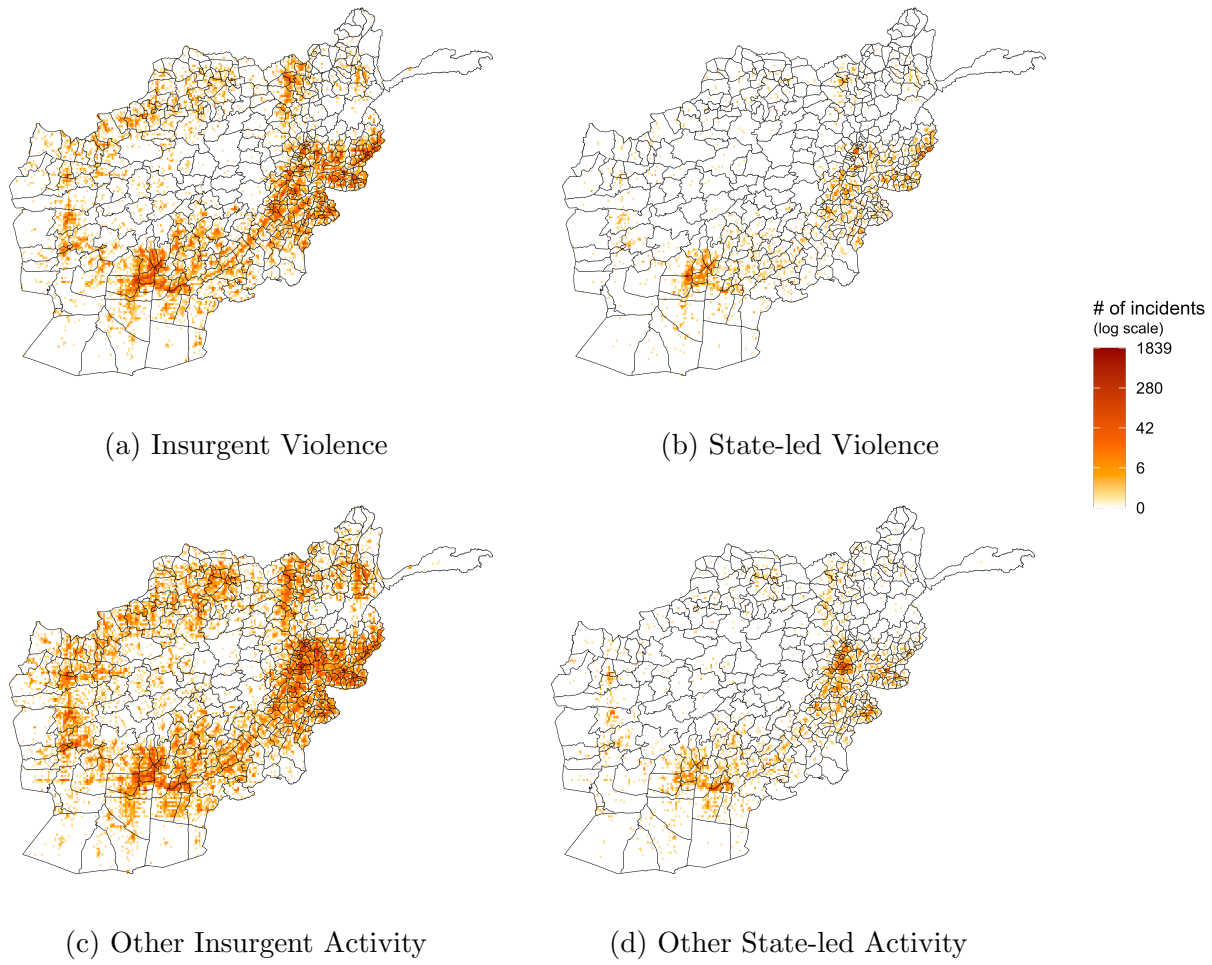
Notes: This figure plots the number of mobile phone users in each month of our data. The total number of users is shown for every month for which we have data for every type of user (phone calls, Data, SMS).

Figure A4: Survey Responses on Religiosity Questions



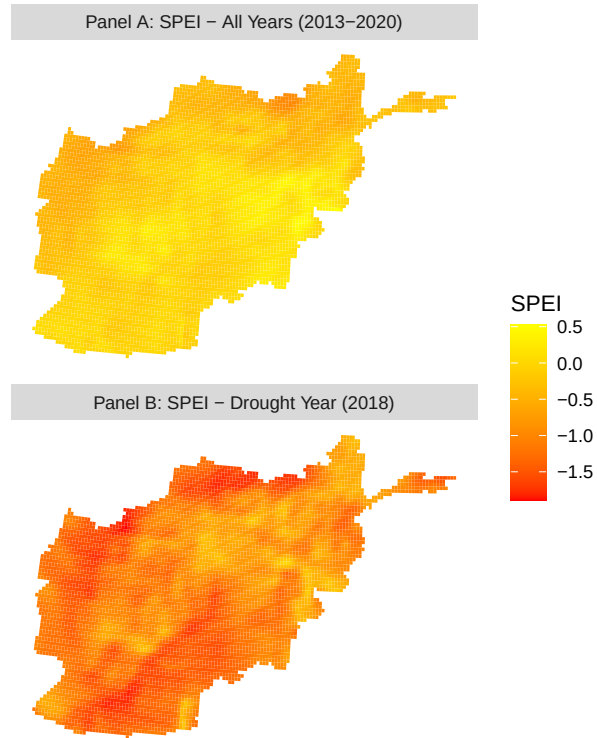
Notes: This figure presents responses to the question “How important are the following practices to you?”. Responses include “Very important”, “Important”, “Moderately important”, “Unimportant” and “Of little importance”.

Figure A5: ISAF Incidents in Afghanistan



Notes: Maps show the location of different types of incidents recorded by the International Security Assistance Force (ISAF) in Afghanistan between 2013-2014. The four categories are (a) insurgent violence, (b) state-led violence, (c) other insurgent activities, and (d) other state-led activities. See footnote ?? for details on these event types. Event counts are aggregated for each 0.05° grid cell. District boundaries are shown in black.

Figure A6: SPEI Measures in Afghanistan



Notes: This figure maps the SPEI measures calculated over the 12 month timescale. Panel A shows the average over the entire sample period. Panel B shows the measures for 2018, a year in which much of Afghanistan experienced a major drought.

Table A1: Comparing the Maghrib Dip in Calls to Shortcodes vs. Standard Numbers

	Maghrib Dip	
	Grid Cell	District
	(1)	(2)
Shortcode Transaction Type (Indicator)	-9.825*** (0.275)	-9.979*** (0.328)
Observations	74954	41986
Mean Non-shortcode Maghrib Dip	15.103	15.855
Grid Cell Fixed Effects	Y	
Month Fixed Effects	Y	Y
District Fixed Effects		Y

Notes: This table examines if the measured Maghrib dip differs statistically based on the transaction type (shortcode calls versus non-shortcode calls). The Maghrib dip is regressed on a indicator of whether the measure uses shortcode calls (versus non-shortcode calls). Column (1) is a grid cell-month level regression which incorporates grid cell and month fixed effects; standard errors, shown in parentheses, are clustered on grid cell. Column (2) is a district-month level regression which uses district and month fixed effects; standard errors, shown in parentheses, are clustered on district. *p<.10, **p<.05, *** p<.01.

Table A2: The Maghrib Dip and Survey Measures of Religiosity

	Maghrib Dip	
	(1)	(2)
Survey Religiosity Index	3.135 (1.913)	3.783** (1.923)
<i>Standardized components:</i>		
Reading the Quran Daily	3.813** (1.678)	4.383*** (1.685)
Observing Namaz 5 Times per Day	0.882 (1.448)	1.414 (1.457)
No Alcohol	3.398* (2.030)	3.866* (2.027)
No Music	3.343* (1.990)	3.363* (2.011)
Fasting during Ramadan	0.980 (1.481)	1.577 (1.496)
Giving Zakat	0.572 (1.446)	1.077 (1.468)
Observations	1092	1092
Mean of Dependent Variable	12.940	12.940
District Fixed Effects	Y	Y
Demographic Controls		Y

Notes: This table examines correlations between the Maghrib dip measure of religious adherence and religious practices self-reported in a household survey. One observation is included for each surveyed individual. Individuals were surveyed in October 2015, so we calculate their Maghrib dip measure using 2015 and 2016 call data. In the top row, the independent variable is an index of responses to survey questions that ask about the importance of six different Islamic religious practices. To construct the index, we convert the responses for each religious practice into an indicator variable (equal to one if the respondent reports the practice is “very important” or “important”), standardize these indicators, and take their average. In the remaining rows, the independent variables are each of the standardized components that comprise the index. All regressions include district fixed effects. Column (2) includes demographic controls for: number of men in their household, number of women in their household, number of children in their household, and an indicator for whether the respondent is of the Hazara ethnic group. Robust standard errors shown in parentheses. *p<.10, ** p<.05, *** p<.01.

Table A3: The Maghrib Dip and Demographic Characteristics of Survey Respondents

	Maghrib Dip
	(1)
Age	0.265* (0.143)
Rural	3.724 (7.804)
Can Read and Write	2.978 (4.161)
Owns Agricultural Land	12.33** (5.609)
Pashtun	13.39** (5.677)
Observations	1043
Mean of Dependent Variable	13.296
District Fixed Effects	Y
Hazara Ethnicity Control	Y

Notes: This table examines correlations between the Maghrib dip measure of religious adherence and demographic characteristics of survey respondents. Since individuals were surveyed in October 2015, we calculate the Maghrib dip calculated using call data over 2015-2016. We regress this Maghrib dip measure on the demographic characteristics shown in the rows of the table. This includes age measured in years; whether the respondent resides in a rural location, can read and write, owns agricultural land and is a member of the Pashtun ethnic group. The regression also includes district fixed effects, and a control for whether the respondent is a member of the Hazara ethnic group. One observation is included for each surveyed individual. Robust standard errors shown in parentheses. *p<.10, ** p<.05, *** p<.01.

Table A4: Ethnicity and the Maghrib Dip

	Maghrib Dip (All Calls)		Maghrib Dip (Shortcode Calls)	
	District	Grid Cell	District	Grid Cell
	(1)	(2)	(3)	(4)
Plurality Pashto Speaking	4.754*** (1.062)	4.903*** (0.957)	3.838** (1.602)	5.552*** (1.505)
Observations	288	595	288	595
Province Fixed Effects	Y	Y	Y	Y
Plurality Hazara Control	Y	Y	Y	Y

Notes: We proxy ethnicity by language spoken in each village. To define the plurality language spoken in each unit (district or grid cell) we aggregate village-level data from 2012 to either district level or grid cell level, restricting to villages within 10km of a cell tower. There are 128 plurality Pashto speaking districts; of the remaining 160 districts, 113 are plurality Dari speaking, 24 Uzbek, 11 Turkmen, 5 Pashai, 4 Balochi, 2 Nuristani, and 1 Hazaragi. Each column is a separate regression. The Maghrib dip is calculated using all calls in columns (1)-(2) and shortcode calls in columns (3)-(4). Columns (1) and (3) are district level regressions in which the district's average Maghrib Dip over 2013-2020 is regressed on an indicator of whether the largest number of villages in the district are plurality Pashto-speaking (while controlling for an indicator of whether the largest number of villages in the district are plurality Hazaragi-speaking). Columns (2) and (4) are grid cell level regressions in which the grid's average Maghrib Dip over 2013-2020 is regressed on an indicator of whether the largest number of villages in the grid are plurality Pashto-speaking (while controlling for an indicator of whether the largest number of villages in the grid are plurality Hazaragi-speaking). All regressions include province fixed effects. Robust standard errors shown in parentheses. *p<.10, **p<.05, *** p<.01.

Table A5: Summary Statistics of Main Variables

	Mean (1)	SD (2)	Min (3)	Max (4)	Obs (5)
A. Variables at District-Month level					
Maghrib Dip	13.006	16.482	−133.907	200.000	4705
Violent Insurgent Events	5.939	12.291	0.000	192.000	4705
Other Insurgent Events	12.254	23.638	0.000	511.000	4705
State-led Violence	0.911	4.391	0.000	99.000	4705
Other State-led Activity	1.145	3.740	0.000	97.000	4705
B. Variables at Grid Cell-Month Level					
Maghrib Dip	15.117	20.910	−200.000	200.000	37496
SPEI	−0.180	0.847	−2.774	2.702	37496
C. Variables at Grid Cell-Year Level					
EVI Max - December	6.292	0.943	−0.736	8.173	4213
Growing Season SPEI	−0.242	0.755	−2.384	1.476	4213
Maghrib Dip in Growing Season	12.728	13.102	−200.000	144.667	3330
Maghrib Dip in Harvest Season	26.443	16.711	−159.788	200.000	3330
Maghrib Dip in Post-harvest Season	10.349	15.419	−200.000	200.000	3330
D. Variables at Grid Cell-Level					
Fraction Rainfed Cropland	0.098	0.191	0.000	0.944	610
Fraction Irrigated Cropland	0.203	0.229	0.000	0.933	610
Fraction Rangeland	0.411	0.306	0.000	0.995	610
Fraction Land Type Aggregated Control	0.260	0.255	0.000	1.000	610
Fraction Built-up	0.029	0.057	0.000	0.608	610

Table A6: Summary Statistics of Additional Variables

	Mean (1)	SD (2)	Min (3)	Max (4)	Obs (5)
A. Variables at Individual Level (In Survey)					
Survey Religiosity Index	-0.000	0.833	-3.792	0.298	1092
Reading the Quran Daily	0.944	0.230	0.000	1.000	1092
Observing Namaz 5 Times per Day	0.953	0.211	0.000	1.000	1092
No Alcohol	0.935	0.247	0.000	1.000	1092
No Music	0.738	0.440	0.000	1.000	1092
Fasting during Ramadan	0.953	0.211	0.000	1.000	1092
Giving Zakat	0.945	0.228	0.000	1.000	1092
Age	39.577	14.049	16.000	87.000	1092
Rural	0.495	0.500	0.000	1.000	1092
Read and Write	0.620	0.486	0.000	1.000	1043
Own Agricultural Land	0.245	0.431	0.000	1.000	1092
Pashtun	0.177	0.382	0.000	1.000	1092
B. Variables at District-Month Level					
Maghrib Dip (Shortcode)	1.244	26.451	-200.000	200.000	4705
Maghrib Dip (Constant Home Location)	13.017	16.961	-122.197	200.000	4488
Maghrib Dip (25 min)	16.081	17.200	-123.348	200.000	4705
Maghrib Dip (35 min)	9.468	15.974	-140.197	200.000	4705
Maghrib Dip (40 min)	6.841	15.819	-143.066	200.000	4705
Maghrib Dip - Before Denom.	11.279	12.586	-100.000	100.000	4704
Maghrib Dip - IHS	2.504	1.760	-5.590	5.991	4705
IHS(Call Volume)	13.164	1.554	6.349	18.553	4705
Taliban Control	0.151	0.359	0.000	1.000	3982
Missing Indicator: Taliban Control	0.154	0.361	0.000	1.000	4705
C. Variables at Grid Cell-Month Level					
Maghrib Dip (Shortcode)	5.277	36.402	-200.000	200.000	37475
Maghrib Dip (25 min)	17.507	21.134	-200.000	200.000	37496
Maghrib Dip (35 min)	12.049	20.753	-200.000	200.000	37496
Maghrib Dip (40 min)	9.645	20.864	-200.000	200.000	37496
Maghrib Dip - Before Denom.	12.801	15.677	-100.000	100.000	37471
Maghrib Dip - IHS	2.562	2.079	-5.991	5.991	37496
IHS(Call Volume)	12.336	1.737	0.881	17.961	37496
D. Variables at District Level					
Plurality Pashto	0.455	0.499	0.000	1.000	398
Plurality Dari	0.392	0.489	0.000	1.000	398
Plurality Neither Dari nor Pashto	0.153	0.361	0.000	1.000	398
District Experienced Any Taliban Control between 2015-2020 (PiX)	0.209	0.407	0.000	1.000	398
E. Variables at Individual-Quarter Level (in CDR)					
Maghrib Dip	10.198	100.014	-200.000	200.000	78419140
SPEI	-0.268	0.861	-2.538	2.033	78419140
F. Variables at District-Year Level					
Poppy - IHS	3.695	3.457	0.000	10.911	1114
Interpolated Poppy - IHS	3.824	3.267	0.000	10.911	1299

Table A7: Addressing Accounts of how Violence affects Religious Adherence

<i>Sample:</i>	Maghrib Dip (2013-2014)	
	Shortcode Calls	Non- Movers
	(1)	(2)
Insurgent Violence	-0.112** (0.047)	-0.068** (0.034)
Observations	4703	4486
District Fixed Effects	Y	Y
Month Fixed Effects	Y	Y
Controls for Other Insurgent Activity and State-led Actions	Y	Y

Notes: In column (1), we restrict to shortcode calls when defining the Maghrib dip. In column (2), we restrict to individuals who have not changed their home location in the past 3 months. We define home location as the district where a caller has made the most calls in a given month. Controls for State-led Actions include the variables “State-led Violence” and “Other State-led Activity”. Standard errors clustered on district shown in parentheses. *p<.10, ** p<.05, *** p<.01.

Table A8: Insurgent Violence and Religious Adherence: Robustness to Additional Controls and Alternate Dip Measures

	Maghrib Dip (2013-2014)							
	(1) (Main)	(2) (Main)	(3) (25 min)	(4) (35 min)	(5) (40 min)	(6) (Before Denom.)	(7) (IHS)	(8) (Main)
Insurgent Violence	-0.073*** (0.028)	-0.081*** (0.030)	-0.054** (0.027)	-0.072** (0.028)	-0.066** (0.029)	-0.076*** (0.025)	-0.013*** (0.004)	-0.072** (0.028)
Maghrib Dip _{t-1}	0.037 (0.044)							
IHS(Total Volume)		-2.723* (1.578)						
Taliban Control								-0.530 (0.899)
Observations	4468	4703	4703	4703	4703	4702	4703	4703
District Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Month Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Controls for Other Insurgent Activity & State-led Actions	Y	Y	Y	Y	Y	Y	Y	Y

Notes: Column (1) controls for the Maghrib dip in the previous month. Column (2) controls for the inverse hyperbolic sine (IHS) transform of the average daily call volume in a month. Columns (3)-(5) construct the outcome by considering the change in call volume 25, 35, and 40 minutes before and after the start of Maghrib, respectively. In column (6), the Maghrib dip is constructed by scaling the change in call volume 30 minutes before and after by the volume 30 minutes before. In column (7), the dependent variable is the IHS transform of the ratio of call volume in the 30 minutes before Maghrib over the average call volume in 30 minutes before and after. Column (8) controls for Taliban control, based on a binary indicator for whether the survey firm (ACSOR) could access the district to conduct the survey (1 if a district was coded as “Totally inaccessible” or “No women – only men can work there”; 0 otherwise), as well as a dummy variable for missing values of Taliban control. Each regression is at the district-month level, using data from April 2013 to December 2014. Controls for State-led Actions include the variables “State-led Violence” and “Other State-led Activity”. Standard errors, clustered by district, are shown in parentheses. *p<.10, ** p<.05, *** p<.01.

Table A9: Insurgent Violence and Religious Adherence: Robustness to Various Samples

<i>Sample:</i>	Maghrib Dip (2013-2014)			
	Non-Ramadan Days	Non-Hazara Districts	# Users (Top 95%)	Call Volume (Top 95%)
	(1)	(2)	(3)	(4)
Insurgent Violence	-0.078*** (0.027)	-0.091*** (0.032)	-0.074*** (0.028)	-0.072** (0.028)
Observations	4603	4693	4511	4514
District Fixed Effects	Y	Y	Y	Y
Month Fixed Effects	Y	Y	Y	Y
Controls for Other Insurgent Activity and State-led Actions	Y	Y	Y	Y

Notes: Column (1) excludes Ramadan days. Column (2) excludes districts in which at least 5% of villages speak primarily Hazaragi. Column (3) excludes the 5% of districts that have the fewest number of active mobile phone subscribers over 2013-2014. Column (4) excludes the 5% of districts with the lowest call volume over 2013-2014. Controls for State-led Actions include the variables “State-led Violence” and “Other State-led Activity”. Standard errors clustered on district shown in parentheses. *p<.10, ** p<.05, *** p<.01.

Table A10: Climate and Religious Adherence: Robustness to Additional Controls & Alternate Dip Measures

	Maghrib Dip							
	(1) (Before Denom.)	(2) (IHS)	(3) (25 min)	(4) (35 min)	(5) (40 min)	(6) (Main)	(7) (Main)	(8) (Main)
SPEI	−0.518*** (0.137)	−0.036** (0.016)	−0.608*** (0.188)	−0.684*** (0.212)	−0.688*** (0.230)	−0.484*** (0.148)	−0.579*** (0.195)	−0.672*** (0.201)
Maghrib Dip _{t-1}						0.189*** (0.031)		
IHS(Total Volume)							−2.038*** (0.356)	
Observations	37470	37496	37496	37496	37496	35362	37496	37496
Grid Cell Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Month Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Clustering on	Grid Cell	Grid Cell	Grid Cell	Grid Cell	Grid Cell	Grid Cell	Grid Cell	District

Notes: In column (1), the Maghrib dip is constructed by scaling the change in call volume 30 minutes before and after by the volume 30 minutes before. In column (2), we take the Inverse hyperbolic sine (IHS) transform of the ratio of call volume in the 30 minutes before over the average call volume in the before and after periods. In columns (3)-(5), we construct the Maghrib dip measure by considering the change in all volume 25, 35, and 40 minutes before and after the start of Maghrib, respectively. Column (6) controls for the Maghrib dip in the previous month. Column (7) controls for the IHS transform of the average daily call volume in a month. In column (8), standard errors are clustered on district, as opposed to on grid. In columns (1)-(7), standard errors clustered on grid cell shown in parentheses. *p<.10, ** p<.05, *** p<.01.

Table A11: Climate and Religious Adherence: Robustness to Various Samples

<i>Sample:</i>	Maghrib Dip					
	Non-Hazara Districts	Non-Ramadan Days	Users (Top 95%)	Call Volume (Top 95%)	Non-Urban Areas	Non-Taliban Districts
	(1)	(2)	(3)	(4)	(5)	(6)
SPEI	−0.705*** (0.203)	−0.523*** (0.195)	−0.632*** (0.197)	−0.563*** (0.191)	−0.726*** (0.233)	−0.753*** (0.220)
Observations	36590	37381	36273	36444	30920	33518
Grid Cell Fixed Effects	Y	Y	Y	Y	Y	Y
Month Fixed Effects	Y	Y	Y	Y	Y	Y

Notes: In column (1), we drop grids in which at least 5% of villages speak Hazaragi. In column (2), we drop Ramadan days. In column (3), we drop the sparsest 5% of grids, defining sparsity based on the number of mobile phone users within the grid cell over 2013-2020. In column (4), we drop the bottom 5% of grids in terms of 2013-2020 call volume. In column (5), we drop the urban areas defined as those in which the fraction of the grid cell with “built up” area exceeds 5%. In column (6), we drop districts under Taliban control, which is directly measured in the PiX data for the 2015-2020 period. Standard errors clustered on grid cell shown in parentheses. *p<.10, ** p<.05, *** p<.01.

Table A12: Climate and Religious Adherence: Addressing Alternate Accounts

	Maghrib Dip	IHS(Call Volume)	Maghrib Dip	
	Shortcode (1)	Full Sample (2)	In Sample Continuously (3)	Non-Movers (4)
SPEI	-1.002*** (0.342)	-0.007 (0.009)	-0.709*** (0.212)	-0.641*** (0.205)
Observations	37475	37496	37141	37138
Grid Cell Fixed Effects	Y	Y	Y	Y
Month Fixed Effects	Y	Y	Y	Y

Notes: In column (1), the dependent variable is the Maghrib dip restricted to shortcode calls. In column (2), the outcome is the IHS transform of the average daily call volume in a month in a grid cell. In columns (3) and (4), the dependent variable is the Maghrib dip across different sample restrictions. In column (3), we restrict the sample to callers who have made at least 1 call in each of the past 3 months. In column (4), in each month, we restrict the sample to users who have only made calls from the same home location over the previous 3 months. We define home location as the district where a caller has made the most calls in a given month. Standard errors clustered on grid cell shown in parentheses. *p<.10, ** p<.05, *** p<.01.

Table A13: Climate and Religious Adherence: Heterogeneity by Initial Religiosity

<i>Baseline religious adherence:</i>	Maghrib Dip (2016–2020)			
	Average		Residual	
	(1)	(2)	(3)	(4)
SPEI	−0.151 (0.136)	−0.125 (0.139)	−0.149 (0.211)	−0.141 (0.212)
SPEI × Above Median Baseline Religious Adherence	−0.127*** (0.042)		−0.128* (0.077)	
SPEI × Tercile 2 Baseline Religious Adherence		−0.043 (0.052)		−0.032 (0.066)
SPEI × Tercile 3 Baseline Religious Adherence		−0.230*** (0.057)		−0.188* (0.100)
Observations	78419140	78419140	78404045	78404045
Individual FE	Y	Y	Y	Y
Grid Cell FE	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y

Notes: Each column is a separate regression. One observation is included for each individual-quarter between 2016 to 2020. SPEI is calculated for each quarter in each individual's home grid, where home is defined as the grid from which they made the most calls in that quarter. The regressions in this table examine heterogeneity by baseline religious adherence. In columns (1) and (2), baseline religious adherence is defined based on the Maghrib dip calculated by aggregating call volumes in the 30-minute windows before and after Maghrib, for all 33 months during the baseline period (spanning 2013-2015). Columns (3) and (4) use a two-stage approach to calculate baseline religiosity as the residual after accounting for SPEI shocks (and grid and quarter fixed effects) during the baseline period (see text for details). The table shows that adverse climate shocks (measured by SPEI) are associated with a larger increase in religious adherence for those who are initially more religiously adherent. Standard errors clustered on grid cell are shown in parentheses. *p<.10, ** p<.05, *** p<.01.

Table A14: Climate and Religious Adherence: Role of Poppy Cultivation

	IHS(Poppy)	IHS(Interpolated Poppy)	Maghrib Dip	
	(1)	(2)	(3)	(4)
SPEI	−0.004 (0.052)	0.009 (0.045)	−0.668*** (0.200)	−0.675*** (0.201)
Observations	20711	24360	37496	37496
Grid Cell Fixed Effects	Y	Y	Y	Y
Month Fixed Effects	Y	Y	Y	Y
Controls – IHS(Poppy)			Y	
Controls – IHS(Int. Poppy)				Y

Notes: In column (1), the dependent variable is the IHS transform of hectares of land cultivated with poppy. In column (2), the dependent variable is the IHS transform of linearly interpolated poppy cultivation. In columns (3) and (4), the dependent variable is the Maghrib dip, and we control for the IHS transform of the poppy cultivation and the IHS transform of linearly interpolated poppy cultivation, respectively. The poppy cultivation variable is at the district-year level. Regressions are at the grid cell-month level, with standard errors clustered on grid cell shown in parentheses. In columns (3) and (4), missing values of the poppy control variable are filled in by an arbitrary number (zero) and an additional control variable is included indicating if missing values have been filled in this way. *p<.10, ** p<.05, *** p<.01.